



8-Phase N-D-C Structural Analysis — Silicon Valley Bank (Q4 2022)

This analysis maps SVB's institutional functions onto 8 structural phases using the N-D-C framework. Each phase is evaluated by its balance between Definition (D: constraints, limits, risk controls in force) and Contribution (C: outputs, deployments, risk exposures produced). The chain average balance is the system-level diagnostic.

The critical finding: Phases 1 and 7 (capital adequacy and regulatory compliance) appear Coherent, matching the regulatory picture. Phases 2, 3, and 4 are in Breakdown, invisible to standard metrics. Phase 3 entered Breakdown in Q3 2021 and remained there for six consecutive quarters before the failure.

61.3%

Chain average balance (π axis)
4 phases below 61.8% threshold

Primary bottleneck: Phase 3 — Interest Rate Risk Management (21.4%). SVB dissolved its interest rate hedges in mid-2021, leaving D (hedging constraint) near zero while C (rate exposure from \$91B HTM bond portfolio) grew with every Fed rate hike. This is the structural origin of the failure cascade.

Phase Balance Scores — Q4 2022 Snapshot

#	PHASE	BALANCE	STATUS	D / C	KEY STRUCTURAL SIGNAL
0	Deposit Franchise Quality Concentration and stability of the funding base	63.6%	STRESSED	D 140 / C 220	93.8% uninsured deposits, single-sector (tech). C >> D: inflows far exceeded concentration controls.
1	Capital Adequacy (Reported) Δ Regulatory capital against risk-weighted assets	97.5%	DECEPTIVE SIGNAL	D 200 / C 195	CET1 12.1% — appeared well-capitalized. Hidden: \$17.7B HTM losses excluded from regulatory capital. True equity was negative.
2	Asset Allocation and Concentration	29.6%	BREAKDOWN	D 80 / C 270	55% of assets in HTM bonds at low yields, average duration 3.7 years. No effective

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	Deployment of deposits into loans, bonds, and cash				concentration or duration limits applied.
3	Interest Rate Risk Management Duration mismatch managed through hedging	21.4%	BREAKDOWN — PRIMARY	D 60 / C 280	All interest rate swaps dissolved in 2021. D (hedging) near zero. C (rate exposure) grew 525 bps in 18 months. Model detected Breakdown in Q3 2021.
4	Liquidity Coverage and Buffer Short-term obligations met through liquid assets	47.4%	BREAKDOWN	D 190 / C 90	Mid-tier bank LCR exemption removed binding constraint. \$16.8B deposit outflows in 2022. HTM bonds illiquid — sale would trigger loss recognition.
5	Earnings and Net Interest Margin Asset yields net of funding costs	81.3%	GOOD	D 160 / C 130	Earnings remained positive but NIM compressed from 2.07% to 1.93% as cost of funds rose faster than locked-in bond yields.
6	Market Confidence and Funding Access Institutional credibility and depositor confidence	57.1%	CRITICAL	D 175 / C 100	Moody's placed on review December 2022. Institutional investors reducing exposure. Equity raise attempt in March 2023 triggered the run.
7	Regulatory Compliance and Reporting ⚠️ Supervisory obligations and examination results	92.5%	DECEPTIVE SIGNAL	D 200 / C 185	CAMELS 2 rating, no enforcement actions. Regulatory framework did not require LCR, excluded AOCI from capital. Supervisors identified concerns but did not escalate.

Technical Dashboard

Intervention Analysis

Resolution Analysis

← Hub